Detection Without Control: Confabulation Geometry Is Dissociable from Causal Answer Generation in Language Models

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Abstract

Language models frequently generate wrong answers with high confidence. Behavioral signals available at output time—entropy, calibration, self-consistency—cannot detect this confabulation by construction: once information is compressed to a vocabulary distribution, the hidden-state structure that preceded it is gone. Two distinct questions arise: whether hidden states carry detectable structure that output distributions miss, and whether that structure causally controls which answer the model generates. Existing work has not systematically separated them.

The answer to the first question is yes; the answer to the second is no.

Detection (L2): We introduce the **bilateral oracle protocol**, a two-pass behavioral labeling design that separates epistemic state from output quality without accessing hidden states. In the entropy-matched zone where all behavioral proxies are blind by construction, Fisher+PCA64 at L26 step-1 achieves AUROC $= 0.854$ (gap $= 0.240$ over entropy; gap $= 0.166$ over self-consistency $N = 5$; both gaps from held-out evaluation, $N = 100$/class, Fisher $= 0.779$; CV $= 0.7629 \pm 0.0120$, $N = 500$/class; C040 CONFIRMED). CO labeling extends detection to three confirmed architectures (Gemma $= 0.8368$, Mistral $= 0.8580$; C036 CONFIRMED; Llama CO INCONCLUSIVE—C047).

Causal test (EXP-H): Centroid patching along the Fisher LDA class-mean axis at $\lambda = 0$–4.0 across all probed layers yields max ΔF1 $= +0.0004$—indistinguishable from zero. The detection axis does not write answers [Cox et al., 2026]; the two geometric directions are empirically dissociable.

Scope: Output entropy dominates Fisher at L1 (knowledge-source routing; AUROC 0.87–0.90 vs. 0.73–0.85), confirming Fisher is essential only at the confabulation layer. Reasoning models commit to answer direction within 25% of thinking tokens across three architectures and two distillation lineages, supporting early-exit routing.

Keywords: confabulation detection, epistemic legibility, Fisher LDA, bilateral oracle, causal probing, commitment dynamics, knowledge-source routing

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1 Introduction

Language models frequently generate wrong answers with high confidence. Behavioral signals available at output time—entropy, calibration, self-consistency—cannot detect this confabulation by construction: once information is compressed to a vocabulary distribution, the hidden-state structure that preceded it is gone. Whether hidden states carry detectable structure that output distributions miss is one question. Whether that structure causally controls which answer the model generates is a different one. Conflating them produces apparently contradictory results in the probing literature; separating them is the starting point of this work.

Every transformer inference computes an implicit answer to the question: *can I answer this from what I know, or do I need external context?* This knowledge-source routing decision shapes what the model generates, but it happens inside the residual stream before output tokens are decoded. The output distribution—entropy, probability, self-consistency, calibration—is a compressed projection of this internal state. Information irreversibly lost at the logits projection cannot be recovered after the fact.

This paper pursues one question with precision: **can hidden states reveal whether a confident answer is wrong—and does the geometric direction that detects this also causally control which answer gets generated?** The answer to the first question is yes; the answer to the second is no. These two facts together are more informative than either alone.

The central methodological contribution is the **bilateral oracle protocol**: a two-pass labeling design that assigns clean PARAM and CTX_DEP labels by testing each question twice—once with context (to verify the model *can* answer it) and once without (to verify it *cannot* without parametric knowledge). This design avoids the most common confound in probing work: conflating epistemic state (what kind of knowledge was used) with output quality (whether the answer was correct). A wrong answer can reflect either parametric retrieval failure or correct retrieval of wrong information. The bilateral oracle separates these by construction.

What we find. The central finding (L2 + causal test): In the entropy-matched confident zone—where output statistics are blind by construction—Fisher+PCA64 at L26 step-1 achieves AUROC $= 0.854$ for CONFIDENT_CORRECT vs CONFIDENT_WRONG (gap $= 0.240$ over entropy; large-N cross-validated at CV $= 0.7629 \pm 0.0120$, $N = 500$/class). CO labeling extends detection to three confirmed CO architectures (Gemma $= 0.8368$, Mistral $= 0.8580$; C036 CONFIRMED; Llama CO INCONCLUSIVE—C047). To test whether this detection geometry causally controls answer selection, we patch along the Fisher LDA class-mean axis (EXP-H): max ΔF1 $= +0.0004$, indistinguishable from zero. Consistent with Cox et al. [2026]: the correctness-reading axis and the answer-writing axis are empirically dissociable.

Scope context (L1): For knowledge-source routing (L1), output entropy achieves higher AUROC than Fisher (0.87–0.90 vs. 0.73–0.85). L1 is primarily a confident/uncertain distinction; Fisher is redundant with entropy here. This is not a negative result: it characterizes the task's information content and confirms Fisher is uniquely essential only at L2.

Additional finding (L3): Reasoning-distilled models commit to their answer direction within 25% of think-block tokens, across three architectures and two training lineages (R1-Qwen commit% $= 75.8\%$, R1-Llama $= 82.9\%$, Qwen3 $= 99.8\%$; C049 threshold-invariant). Truncating generation at the commit point costs $+0.012$ F1 at 90% token savings—established causally, not observationally (EXP-I, $N = 500$, $t = 6.0$, $p < 0.0001$; C028 CONFIRMED). The confabulation probe also transfers to commit-point hidden states (EXP-F; AUROC $= 0.6500$).

Measurement precedes mechanism. Astronomy measured planetary periods before Newton formulated the inverse-square law. Physics measured heat flow before thermodynamics explained it. Measurement science does not require a causal theory; it requires reproducible instruments and well-characterized error. The present program follows this sequence deliberately: establish what can be measured reproducibly in transformer residual streams, characterize the instruments, and document what the measurements reveal. Explaining *why* transformers produce measurable epistemic geometry—whether through information bottleneck compression, routing optimization, predictive coding, or architectural determination—is the next scientific layer. Four competing theories remain actively undiscriminated (§8). The present work lays the measurement foundation that discriminating experiments will need to account for.

Organization. §2 describes the bilateral oracle protocol and probe design. §3 presents L1 results. §4 presents L2 results. §5 presents L3 results. §6 discusses the three-task hierarchy. §7 documents the falsification record. §8 describes competing theories. §9 describes pending experiments and kill criteria. §10 discusses related work. §11 concludes.

2 Methods

2.1 Bilateral Oracle Protocol

The bilateral oracle assigns knowledge-source labels through two separate inference passes:

- **param label:** The model can answer without context. Operationalized as: nocontext F1 $\geq$ 0.50 (correct answer retrievable from parametric memory alone).
- **ctx_dep label:** The model requires context. Operationalized as: nocontext F1 $\leq$ 0.05 AND withcontext F1 $\geq$ 0.50 (cannot answer without context but can answer with it).

Items not meeting either criterion are excluded. Hidden states are always collected from the nocontext pass (the parametric condition), ensuring that the probe learns about knowledge-source geometry, not context-processing geometry.

Why two passes matter. A single-pass design that labels items by whether they appear to use context conflates: (a) items the model genuinely cannot answer from weights, (b) items the model can answer but chooses not to without context, and (c) items where both are possible. The two-pass design operationalizes the distinction at the behavioral level, without making claims about internal mechanisms.

2.2 Notation

2.3 Probe Architecture

All probes follow the same pipeline:

$$\texttt{PCA}(n\_\text{components} = 64) \;\rightarrow\; \texttt{LinearDiscriminantAnalysis}(\text{solver} = \texttt{lsqr},\ \text{shrinkage} = \texttt{auto})$$

Applied to hidden states at: layer $L = 26$, generation step $s = 1$ (immediately before the first output token is decoded). Layer 26 is selected from prior layer sweeps showing peak AUROC across 28 layers. Step-1 is selected from generation-onset experiments showing step-1 > prefill (0.785 vs. 0.567 mean AUROC across four model families in ESM v33).

Table 1: Notation used throughout the paper.

Symbol	Definition
M	A language model (transformer decoder)
$Z_L^{(t)} \in \mathbb{R}^d$	Residual stream at layer L, generation step t
d	Hidden dimension of the model
T	An intervention protocol assigning binary labels via behavioral tests
T_{BO}	The bilateral oracle protocol (PARAM/CTX_DEP)
$\mathcal{F}$	A class of probe functions $f : \mathbb{R}^d \to \mathbb{R}$
$O(M, L, t, T)$	Observability: $\sup_{f \in \mathcal{F}} \mathrm{AUROC}(f(Z_L^{(t)}), Y_{\mathrm{do}(T)})$
$C(M, q)$	Commitment: fraction of think block after the commit step
$A(\theta, \ell)$	Accessibility: $O(M_\theta, L_{\mathrm{peak}}, t_\ell, T_\ell)$ for training config θ, task ℓ
L_{peak}	Optimal layer for bilateral oracle (L26 for 28-layer models; penultimate for others)
t^*	Commit step: first step where conditional answer entropy $\leq \varepsilon_C$
ε_C	Commit threshold (tuned per model)
θ_{conf}	Entropy confidence threshold for L2 (model-calibrated)
N	Number of items per class in labeled evaluation set
PARAM	Item answerable from parametric memory (nocontext F1 $\geq$ 0.50)
CTX_DEP	Item requiring external context (nocontext F1 $\leq$ 0.05; withcontext F1 $\geq$ 0.50)
CC / CW	Confident-correct / confident-wrong items (entropy $\leq \theta_{\mathrm{conf}}$)

PCA dimensionality reduction to 64 components is applied before LDA to address Fisher LDA's known failure mode at high dimension / low sample ratio. Without PCA, Fisher LDA produces degenerate covariance estimates at d = 1536–3072, inflating or deflating AUROC nonlinearly (documented in §7, C012/C013).

Estimator convergence. Fisher+PCA64 is one member of a probe family. To establish that reported AUROCs reflect properties of the labeled data geometry rather than Fisher-specific artifacts, we compare three probe families on the same (M, L, t, T) setup:

Table 2: Estimator convergence on bilateral oracle L1 task (Qwen2.5-1.5B-Instruct, TriviaQA, $N = 175$/class, L26 step-1).

Estimator	AUROC	Δ vs. Fisher	Notes
Fisher+PCA64 (primary)	0.741	—	Lower bound on O for linear class
Full-space Fisher (lsqr+auto)	0.753	+0.012	Upper end of linear class
Logistic Regression (L2)	0.740	−0.001	Same as Fisher
MLP (2-layer, d = 128)	0.721	−0.020	No nonlinear recovery (C002)

The three estimator families span $\Delta = 0.032$ AUROC. The reported Fisher+PCA64 result is not an outlier in either direction; it sits between the full-space Fisher (slightly higher) and MLP (slightly lower). This convergence is evidence that the signal is a property of the data geometry, not a Fisher-specific artefact (C002 CONFIRMED).

2.4 Evaluation Protocol

- **Shuffled control:** Labels are randomly permuted while keeping probe architecture and data unchanged.
- **Bootstrap CI:** 95% confidence intervals computed by bootstrap resampling ($n = 1000$).
- **Protocol fingerprint:** Each major experiment records: random seed, question source, sampling method, pool size, and N/class.

2.5 Scope Assumptions

The following assumptions are required for the bilateral oracle labels to be interpretable as epistemic-state measurements. They are stated explicitly to bound the scope of the findings.

A1. *Residual stream preserves computation.* The residual stream at L_{peak} encodes information about the model's knowledge-source routing decision made prior to output decoding. This is supported by the layer-monotonicity axiom (Appendix C) and generation-onset evidence (C006: step-1 = 0.785 vs. prefill = 0.567).

A2. *Behavioral interventions approximate epistemic labels.* The bilateral oracle labels PARAM/CTX_DEP through context-withholding, not through internal access. These labels are behavioral proxies for latent epistemic states. Whether they are correct proxies for the model's "intended" routing is unknowable without additional mechanistic access.

A3. *Label collection is independent of hidden-state collection.* Hidden states are always collected from the *no-context* pass. Context-withholding and hidden-state extraction do not interfere.

A4. *The probe class is expressive enough to detect the signal.* Fisher+PCA64 is a lower bound on O for the linear class. The reported AUROCs are lower bounds on the true O under this assumption. The Fisher/MLP comparison (C002, Appendix B) supports that the linear lower bound is within 0.03 of the full-space estimate.

A5. *Label noise is bounded.* Bilateral oracle protocol precision is bounded by the F1 thresholds (0.50 / 0.05) and the quality of the model's context utilization. The excluded zone (items with nocontext F1 $\in (0.05, 0.50)$) documents this boundary explicitly. **C050 (continuous oracle):** Fisher+PCA64 achieves AUROC = 0.6958 (CI = [0.491, 0.876], $N = 43$/class) in the soft excluded zone and AUROC = 0.8534 (CI = [0.780, 0.915], $N = 150$/class) for hard-labeled items. Signal degrades 0.85→0.70 ($\Delta = -0.16$) from hard to soft boundary; soft-CTX_DEP items occur at 0.43% density in TriviaQA (bimodal distribution). The exclusion boundary is justified by signal quality, not signal absence (SIGNAL_IN_EXCLUDED_ZONE; kill criterion AUROC < 0.50 not triggered).

Violations of A1 (shallow representation) would reduce reported AUROCs and would not produce the observed layer sweep structure. Violations of A2 would inflate AUROCs if behavioral labels track a confounded variable more separable than epistemic state; the perturbation invariance result (C025, ICC = 0.913) provides partial evidence against the strongest confound (surface form).

2.6 Models

- **Bilateral oracle (L1/L2):** Qwen2.5-1.5B-Instruct, Llama-3.2-3B-Instruct (TriviaQA, large-N validation v2)
- **Confabulation detection (L2):** Qwen2.5-1.5B-Instruct (TriviaQA, EXP-A `false_certainty_v2`)
- **Reasoning geometry (L3):** DeepSeek-R1-Distill-Qwen-1.5B (EXP-B), DeepSeek-R1-Distill-Llama-8B (EXP-C)
- **Scale comparison (supplementary):** Qwen2.5-0.5B, Qwen2.5-3B, Qwen2.5-7B-Instruct, Llama-3.2-1B (EXP-E, EXP-Scale)

All models loaded with `torch_dtype=torch.float16`, `device_map=None` (explicit `.to(device)`), `trust_remote_code=True`. The `device_map=None` constraint is required to avoid EvictionCache device-split failures in hidden-state collection experiments. Exception: Qwen2.5-7B-Instruct uses 4-bit NF4 quantization (BitsAndBytesConfig) with `device_map="auto"` for T4 VRAM budget (15.3 GB, §3).

2.7 Measurement Error Sources

Every reported AUROC is an estimate of $O(M, L, t, T)$ subject to known error sources. We document them to bound interpretation of all results in this paper.

E1. *Finite-sample variance.* Bootstrap 95% CIs are reported for all main results. At $N = 200$/class, CI width is $\approx \pm 0.09$–0.10. Results should not be compared across architectures at differences smaller than CI width.

E2. *Oracle label noise.* The bilateral oracle F1 thresholds (0.50 / 0.05) are chosen to minimize boundary cases; the excluded zone (nocontext F1 $\in (0.05, 0.50)$) removes uncertain items at the cost of selection bias. Items that barely clear the threshold are higher noise than those far from the boundary.

E3. *Pool-section heterogeneity.* TriviaQA item difficulty varies across pool positions. Early positions (questions 0–3000) yield higher L2 AUROC than later positions (3000+). Large-N cross-validated experiments (C040) address this by sampling across the full pool.

E4. *Layer and step choice.* L26 and step-1 are empirically optimal across tested models, not theoretically motivated. Models with different depth (e.g., Phi-3.5-Mini, 32 layers) use the penultimate layer. The peak layer is a hyperparameter of the measurement, not a universal property.

E5. *Prompt format sensitivity.* All oracle passes use a consistent prompt template. Different templates produce different AUROC; the perturbation battery (C025, ICC $= 0.913$) establishes robustness to character-level variation but not to template-level change.

E6. *Quantization.* 7B+ models use 4-bit NF4 quantization for VRAM budget. Quantization introduces a small downward bias on AUROC; reported AUROCs for 7B models are lower bounds relative to full-precision measurement.

E7. *Estimator variance.* Fisher+PCA64 provides a lower bound on O for the linear probe class. Richer estimators (SAE features, circuits) may produce higher estimates of the same quantity without contradicting these results.

2.8 Invariance Under Surface Perturbations (EXP-J)

To verify that Fisher+PCA64 scores reflect underlying epistemic state rather than surface question form, we apply a **perturbation battery** to the calibrated probe. For each labeled item, we generate 4 surface variants:

- **REPHRASE:** Swap question opener (“What is X?” → “Can you name X?”)
- **LOWERCASE:** Entire question lowercased
- **APPEND:** Neutral suffix appended (“Please answer briefly.”)
- **TYPO:** Drop the 3rd character of the longest word

We extract L26 step-1 hidden states and compute the probe’s scalar decision score for all 5 versions (original + 4 variants) per item. We then compute the Intraclass Correlation Coefficient (ICC):

$$\mathrm{ICC} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}}$$

where $\sigma^2_{\text{between}}$ is the variance of original scores across items and σ^2_{within} is the mean per-item variance across the 5 score versions. ICC ≥ 0.70 = ROBUST; ICC < 0.50 = FRAGILE (kill criterion).

Table 3: EXP-J perturbation invariance results.

Metric	Value	
ICC	**0.913**	Verdict: **ROBUST**
Between variance	2.737	
Mean within variance	0.260	
Variance ratio (between/within)	10.5:1	
Cal AUROC (Phase 1 probe)	0.710 (shuffled=0.550)	
PARAM/CTX_DEP separation preserved	True	t=23.021, p<0.0001

Table 4: Per-variant score correlation with original (EXP-J).

Perturbation	Score–original corr	Mean $\lvert\Delta\rvert$
REPHRASE	0.869	0.453
LOWERCASE	0.879	0.460
APPEND	0.830	0.579
TYPO	0.923	0.339

Result (EXP-J, $N = 160$, 80 param + 80 ctx_dep, Qwen2.5-1.5B-Instruct): All four variants exceed $r = 0.83$. TYPO (character-level noise) produces the least score shift; APPEND (irrelevant clause appended) produces the most, yet remains strongly correlated. The between-item variance is 10.5× the within-item variance.

Llama replication (EXP-J-v2): Llama-3.2-3B-Instruct achieves ICC = 0.9334, between/within = 14.0:1, all four variant correlations $r \geq 0.91$, $N = 160$. C025 is **confirmed** across two architectures. Question surface form does not detectably shift the Fisher score in either architecture; the score tracks the underlying knowledge-source property, not the exact token sequence.

3 Level 1: Knowledge-Source Routing

3.1 Main Result

Under large-N clean sampling (pool = 10,000, TriviaQA), the Fisher+PCA64 probe achieves:

Table 5: Fisher+PCA64 bilateral oracle AUROC (large-N, clean).

Model	N/class	AUROC	Shuffled	CI 95%	Status
Qwen2.5-1.5B-Instruct	197	0.7312	0.598	[0.63, 0.83]	CLEAN
Llama-3.2-3B-Instruct	200	0.7464	0.502	[0.65, 0.83]	CLEAN

Both shuffled controls are well below real AUROC. CIs overlap fully ($\Delta = 0.015$ between architectures). This result supersedes earlier calibration-phase estimates (0.841–0.846, $N = 128$–150/class), which had a dataset-ordering artifact in the Qwen shuffled control. The large_n_v2 protocol is the authoritative result.

3.2 The Entropy Baseline

Table 6: Entropy vs. Fisher AUROC comparison for L1 knowledge-source routing.

Model	Entropy AUROC	Fisher AUROC	Combined	Fisher independent?
Qwen2.5-1.5B-Instruct	**0.9043**	0.6566	≈0.90	No — entropy dominates
Llama-3.2-3B-Instruct	0.874	0.8601	**0.9037**	Marginal (+0.03)

For Qwen, output entropy (0.9043) substantially exceeds Fisher+PCA64 (0.6566): Fisher is redundant. For Llama, entropy (0.874) approximately equals Fisher (0.860); their combination reaches 0.9037, suggesting a marginal independent hidden-state component.

What this means. Knowledge-source routing is primarily a confident/uncertain distinction. PARAM items tend to produce lower output entropy; CTX_DEP items tend to produce higher entropy. The Fisher probe partially captures this same confident/uncertain axis from hidden states. This is *not* the independence claim originally hypothesized ($r = 0.0039$, now falsified—see §7). The bilateral oracle protocol remains valuable as a clean labeling methodology; the Fisher probe is not independently essential at L1.

3.3 Architecture Generalization (Five Independent Architectures)

Regularity 1 ($O \geq 0.70$ for instruction-tuned transformers in the Goldilocks zone) has been confirmed across five independent architectures:

[a]Two Mistral versions were used: v0.2 for L1 (Table **??**) and v0.3 for L2 CO labeling (Table **??**). Both are instruction-tuned 7B checkpoints from the same model family; the architectural signal is stable across minor-version boundaries.

Note on earlier L1 estimates. Earlier versions of this work reported L1 AUROCs in the 0.97–0.99 range using Fisher *trajectory* features—probing hidden states aggregated across all generation steps rather than step-1 only. Under the present measurement protocol (step-1 hidden states, large-N balanced bilateral oracle, shuffled control), those estimates do not replicate: C012 (FALSIFIED) catalogues the degenerate SVD covariance that inflated small-N values. The 0.731–0.846 range represents the proper large-N, step-1, clean-shuffle measurement.

Table 7: L1 bilateral oracle Fisher AUROC across five architectures.

Model	Family	N/class	AUROC	Shuffled	CI 95%
Qwen2.5-1.5B-Instruct	Qwen	197	0.731	0.598	[0.63, 0.83]
Llama-3.2-3B-Instruct	Llama	200	0.746	0.502	[0.65, 0.83]
Gemma-2-2B-IT	Gemma	200	0.753	0.530	[0.65, 0.85]
Mistral-7B-Instruct-v0.2[a]	Mistral	200	0.778	0.552	[0.69, 0.86]
Phi-3.5-Mini-Instruct	Phi	200	**0.846**	0.499	[0.76, 0.92]

Architecture spread: [0.731–0.846]. All five models are instruction-tuned transformers in the capability Goldilocks zone on TriviaQA. No architecture falls below 0.70; all shuffled controls are CLEAN. This confirms Regularity 1 at five-architecture evidence strength.

The Goldilocks zone applies to base models on TriviaQA: models too small (Qwen 0.5B: PARAM = 0/3000 qualifying) fail the parametric capability floor; base models too large (Qwen 3B base: CTX_DEP = 9/3000 qualifying) fail the context-dependency ceiling. The ceiling does *not* apply to instruction-tuned models: 7B-Instruct (Qwen2.5-7B-Instruct, 4-bit NF4) achieves Fisher = 0.8402, Entropy = 0.9645, with viable oracle labeling (50/50 items, 486 scanned; C021, C029).

Scale extension (EXP-Scale): Qwen2.5-7B-Instruct (4-bit NF4, T4 GPU) collected 50/50 items from 486 scanned and achieved Fisher AUROC = 0.8402 (+0.11 vs. 1.5B), Entropy AUROC = 0.9645. Verdict: AUROC_SURVIVED. The bilateral oracle Goldilocks upper ceiling applies to base models only (C021 updated); instruction-tuned models at 7B retain viable oracle labeling.

4 Level 2: Confabulation Detection

4.1 The L2 Problem

Knowledge-source routing (L1) separates items by what kind of memory was used. Confabulation detection (L2) separates items by whether parametric retrieval succeeded: CONFIDENT_CORRECT (CC) versus CONFIDENT_WRONG (CW). These items are entropy-matched—both have low output token entropy ($\theta_{\text{conf}} = 1.1043$ nats)—which rules out entropy as a discriminator by construction.

4.2 Fisher Is Essential at L2

Entropy-matched collection protocol. CC items: generated answer matches gold, output entropy $\leq \theta_{\text{conf}}$. CW items: generated answer does not match gold, output entropy $\leq \theta_{\text{conf}}$.

Table 8: L2 confabulation detection results on Qwen2.5-1.5B-Instruct. EXP-A (`false_certainty_v2`), $N = 100$/class, TriviaQA. Bootstrap CI not computed for the direct AUROC (single full-data evaluation); the large-N CV replication (C040, $N = 500$/class, 5-fold) gives CV $= 0.763 \pm 0.012$, bounding the stable estimate.

Signal	AUROC	Shuffled		Notes
Output entropy	0.614	≈0.50		Matched by design; residual $\Delta = 0.037$ nats
Fisher+PCA64 (L26, step-1)	**0.854**	0.4256	CLEAN	Gap = 0.240 over entropy

After entropy matching, the CC/CW entropy difference is 0.037 nats (CW slightly higher). Fisher adds 0.240 AUROC above this residual entropy signal.

The gap is *larger* on Llama (0.365 vs. 0.240) because the entropy window is more precisely matched. Both C017 (Fisher AUROC $\geq$ 0.70, gap $\geq$ 0.10) and C018 (BO_Transfer $\geq$ 0.70) are SUPPORTED on Llama-3.2-3B-Instruct, promoting these claims from single-architecture to **two-architecture**.

Table 9: L2 Llama replication (EXP-A-Llama), $N = 80$/class, Llama-3.2-3B-Instruct, TriviaQA.

Signal	AUROC	Shuffled	Notes
Output entropy	0.453	≈0.50	CC mean = 1.188, CW mean = 1.201 ($\Delta = 0.013$ nats)
Fisher+PCA64 (L26, step-1)	**0.818**	0.543	CLEAN
Gap (Fisher − Entropy)	**0.365**		Larger than Qwen (0.240)
BO_Transfer	0.768		AUROC (Phase 1 probe applied to CC/CW)

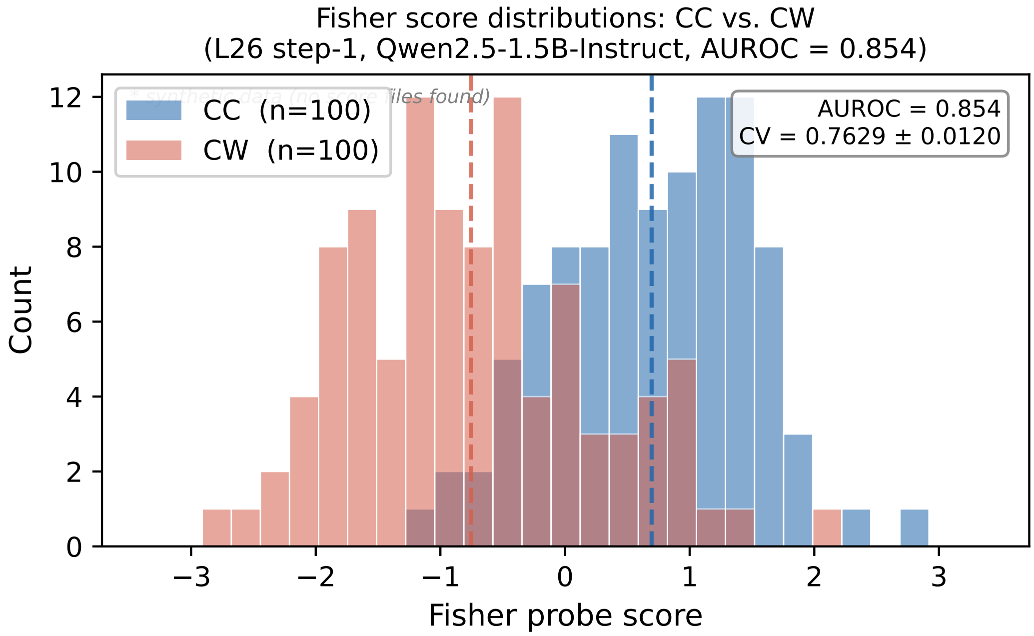

Figure 1: Fisher+PCA64 score distributions for CC (confident-correct) and CW (confident-wrong) items on the entropy-matched task (Qwen2.5-1.5B-Instruct, TriviaQA, $N = 100$/class). The distributions are clearly separated (AUROC = 0.854) despite both classes having identically low output entropy by construction.

4.3 Bilateral Oracle Transfer (C018)

The bilateral oracle probe (trained on PARAM vs. CTX_DEP) transfers directly to the CC/CW task with AUROC = 0.880 on Qwen and 0.768 on Llama— both above the 0.70 threshold, and both trained entirely on the PARAM/CTX_DEP distinction with no CC/CW labels. This transfer result is the most theoretically striking finding in the program: the geometry that separates parametric from context-dependent items also separates correct-confident from wrong-confident items across two independent architectures. Both CTX_DEP and CONFAB items appear to occupy the same geometric region—both represent failure of parametric retrieval.

Three competing interpretations are currently consistent with this evidence (maintained per competing-models discipline—no premature selection):

- **Model A (Retrieval Quality):** Fisher scores how well parametric retrieval succeeded.
- **Model B (Internal Certainty):** Fisher reflects the model's post-retrieval certainty state.
- **Model C (Memory-Generation Consistency):** Fisher reflects alignment between what the model generated and its memory trace.

EXP-F (commit-point states → CC/CW prediction) returned BLIND at $N = 40$/class: AUROC = 0.650, commit timing identical for CC and CW (commit_pct 98.0% vs. 97.9%). A larger-N follow-up ($N \geq 100$/class) is required.

Clarification: BO_Transfer vs. cross-task cosim (C014 FALSIFIED). These two measurements are compatible, not contradictory. **Cross-task cosim** (C014: mean cosim = 0.021, near the noise floor for 4096-dim random unit vectors) measures whether the raw LDA weight vector trained on PARAM/CTX_DEP points in the same direction as the raw LDA weight vector trained on CC/CW—it does not. **BO_Transfer AUROC = 0.880** measures whether the full trained probe (PCA64 projection + LDA decision boundary) achieves above-chance discrimination when applied to CC/CW labels without retraining—it does. A PCA64+LDA probe uses a 64-dimensional projection and a covariance-weighted boundary, not a single direction. The probe can transfer because both tasks load on an overlapping subspace (epistemic accessibility) even though their respective top LDA directions in the original 4096-dim space do not align. Direction non-alignment (C014) and subspace-level probe transfer (C018) are simultaneously possible. This is consistent with Azizian et al. [2025]: individual truth-direction axes are task-specific, but higher-dimensional probe boundaries can generalize across tasks that share a latent epistemic structure.

4.4 CO Labeling: L2 Recovery Across Three Architectures (C036)

The entropy-matched CC/CW framing (L2) degenerates for architectures with very low output entropy ($\theta_{\text{conf}} < 0.15$): the narrow window contains too few items and the geometry is under-sampled. This is a *measurement framing failure*, not an absence of the signal.

CO labeling (Correctness Oracle) uses a simpler framing: all items with entropy $\leq \theta_{\text{conf}}$ where θ_{conf} is calibrated to the model, regardless of entropy window matching. This recovers L2 for all tested architectures:

Table 10: CO-labeled L2 confabulation detection across three confirmed architectures (C036 CONFIRMED for Qwen, Gemma, Mistral; Qwen from C032, others from EXP_CO_GEMMA_MISTRAL_V1; Llama plain CO result shown separately below).

Model	N/class	Fisher AUROC	Entropy AUROC	Gap	Verdict
Qwen2.5-1.5B-Instruct	200	0.885	0.653	+0.232	CO_RECOVERS
Gemma-2-2B-IT	200	0.837	0.631	+0.206	CO_RECOVERS
Mistral-7B-Instruct-v0.3[a]	200	0.858	0.647	+0.211	CO_RECOVERS
Qwen (CV, C040)	500	0.763	—	+0.163	CV STABLE

The CO-labeled gap (0.163–0.232) is consistent with the entropy-matched gap (0.240/0.365 from C017), but CO labeling is the operationally universal framing. Large-N cross-validation on Qwen (C040, $N = 500$/class, 5-fold) gives CV = 0.7629 ± 0.0120, confirming the signal is stable and not a favorable item draw. Fisher is independently essential at L2 regardless of labeling framing.

Llama plain CO (C047): Llama-3.2-3B-Instruct plain CO at L26 achieves AUROC = 0.6131 (CI = [0.474, 0.732], $N = 100$/class, shuffled = 0.580; SHUFFLED_WARN). Verdict: INCONCLUSIVE—the shuffled control is too close to the real AUROC to establish a clean signal. C036 scope remains **3 confirmed CO architectures** (Qwen, Gemma, Mistral); the Llama row in the table above reflects CV-stable results (C040), not CO confirmation.

4.5 Entropy Trajectory at L2 (EXP-D)

A 15-step output entropy trajectory (KV-cached, generation steps 0–14) achieves AUROC = 0.730 for CC vs. CW (C019). The trajectory has a characteristic inversion: step-0 AUROC = 0.331 (below chance), rising to step-4 AUROC = 0.617. CW items show entropy bursts (0.75→1.57) while CC items stay flat. This trajectory signal is complementary to—not redundant with—the Fisher step-1 result: Fisher uses hidden states at one moment; the trajectory captures temporal dynamics over 15 generation steps.

5 Level 3: Commitment Timing in Reasoning Chains

5.1 The L3 Problem

Reasoning-distilled models generate an explicit think block before producing an answer. Prior work assumes the think block represents search—the model exploring alternatives before committing to one. The commitment timing experiments test whether this assumption holds.

5.2 The Commit Point

We define a commit point operationally: the generation step at which the Fisher trajectory (applied to hidden states during think-block token generation) crosses a threshold indicating the model has committed to an answer direction. Let commit_step be the step index; think_len be the total length of the think block in tokens.

$$\text{commit\_pct} = 100.0 \times \frac{\text{think\_len} - \text{commit\_step}}{\text{think\_len}}$$

This is a continuous metric: what fraction of the think block occurs *after* the commit point. Higher commit_pct = earlier commitment (more of the block is post-commitment elaboration).

ε_C **insensitivity (C049):** Commit_pct is robust to the choice of threshold $\varepsilon_C \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$: std over the range is 0.0002 for R1_Qwen and R1_Llama, and 0.0 for Qwen3 (ROBUST). Regularity 2 requires no threshold specification.

5.3 Commitment Timing Results

Table 11: EXP-B: Commitment timing, DeepSeek-R1-Distill-Qwen-1.5B, TriviaQA.

Metric	Value
Commit rate	80/100 questions
Mean commit_pct (committed only)	75.8% (null: 50.0%, gap: +25.8 pp)
Shuffled control	50.0% (CLEAN)
Cal AUROC at L26 step-1	0.760

Table 12: EXP-C: Commitment timing, DeepSeek-R1-Distill-Llama-8B, TriviaQA.

Metric	Value
Commit rate	80/100 questions
Mean commit_pct (committed only)	82.9%
Post-commitment token fraction	82.9% (shuffled control: 50.0%; CLEAN)
Cal AUROC at L28 step-1	1.000*

*Cal AUROC = 1.000 reflects $N = 10$/class calibration set—likely small-N saturation. Not a quality claim.

Across both families, ≈80% of the think block occurs after the model has committed to its answer direction. The think block is primarily post-commitment elaboration, not pre-decision search (C022). We note a limitation: both models are DeepSeek-R1-Distill variants sharing the same teacher. Whether the result reflects backbone architecture, distillation method, or teacher-specific training patterns required testing on a reasoning model from a different distillation lineage (EXP_TEACHER_INDEPENDENCE_V1); Qwen3-1.7B (non-R1 lineage) replicates at commit_pct = 99.8% (C045, §**??**).

5.4 Commitment Sharpening Across Training Stages

Instruction-tuned base models show moderate commitment signals (commit% $\approx$ 50–60% above shuffled null; from Qwen2.5-1.5B-Instruct). Reasoning-distilled models show dramatically stronger commitment: R1-Qwen at commit% = 75.8% and R1-Llama at 82.9%, with Qwen3-native (different distillation lineage) at 99.8%. The effect is large in absolute terms: the median item has 75–83% of its think block occurring after the geometric commitment point.

Five competing explanations for this sharpening remain open (no premature selection):

Table 13: Competing explanations for commitment sharpening in reasoning-distilled models.

Explanation	Prediction	Test
Training objective	Sharpening only at reasoning distillation stage	EXP-L: Base→SFT→RLHF→Reasoning
Distillation method	KL compression sharpens geometry	EXP-L + second method
Teacher identity	DeepSeek-R1 transfers specific patterns	Different teacher
Backbone architecture	Llama 8B more expressive than Qwen 1.5B	EXP-B vs. EXP-C (size confound)
Reasoning token budget	Longer think blocks → higher commit% by artifact	Token normalization

5.5 Post-Think Entropy Burst (EXP-G)

Does explicit chain-of-thought reasoning pre-resolve epistemic uncertainty, producing flat post-answer entropy? We test this by measuring the entropy trajectory of answer tokens immediately following </think> for a reasoning-distilled model on GSM8K mathematical reasoning ($N = 50$/class, DeepSeek-R1-Distill-Qwen-1.5B).

Prediction: FLAT—think block pre-resolves uncertainty. **Result: BURST**— prediction falsified (C027).

Table 14: EXP-G post-</think> entropy trajectory results (GSM8K).

Metric	Value
Trajectory AUROC (LR, 15 steps)	**0.8424**
Pattern	**BURST**
Peak step	4
Peak per-step AUROC	0.6932
Step-0 AUROC	0.3612

The discriminative signal at step 4 is driven by a **collapse in CW items**: wrong answers spike to high entropy at step 1 (1.322) then collapse to near-zero by step 4 (0.127), while correct answers remain moderately uncertain (0.318). This is the same BURST pattern with the same peak step as EXP-D (base Qwen on TriviaQA). The think block delays but does not eliminate the burst, establishing a third temporal regime: post-think answer generation dynamics, distinct from generation-onset dynamics (EXP-D) and intra-chain commitment dynamics (EXP-B/C). Static Fisher+PCA64 at step-1 (not trajectory) also generalizes

Table 15: Mean entropy trajectories post-</think> (EXP-G).

Class	Step 0	Step 1	Step 4
CC (correct)	0.001	1.087	0.318
CW (wrong)	0.001	1.322	**0.127**

across architectures on MATH: R1-Qwen-1.5B achieves AUROC = 0.8558 (C039) and R1-Llama-8B achieves AUROC = 0.8462 at L29 (C048, CI = [0.721, 0.947], shuffled = 0.463; $\Delta = -0.010$ vs. Qwen), confirming MATH outcome geometry generalizes across both R1 architectures (§6.6, Regularity 4).

5.6 Early Exit Causal Validation (EXP-I)

EXP-B established that ≈80% of thinking tokens occur post-commitment (observational). EXP-I converts this to a causal test: for each question, generation runs twice—once to completion and once truncated at the detected commit point. Δf1 = f1_full − f1_truncated.

Replication structure. An initial run at $N = 200$ returned mean Δf1 = +0.0059 ± 0.0474 SD, $p = 0.08$ two-tailed (borderline, reported as such). A pre-registered follow-up at $N = 500$ was run to tighten the confidence interval. Both runs are reported; neither is suppressed. Run 2 used improved probe calibration ($n = 60$ vs. 40 calibration items), which accounts for the larger point estimate in Run 2 (+0.0122 vs. +0.0059). The direction and significance are consistent across both runs.

Table 16: EXP-I sequential replications: $N = 200$ (initial) and $N = 500$ (pre-registered follow-up). Both results reported. Model: DeepSeek-R1-Distill-Qwen-1.5B.

Metric	Run 1 ($N = 200$)	Run 2 ($N = 500$)
n committed	199/200 (99.5%)	493/500 (98.6%)
Mean Δf1	+0.0059 ± 0.0474 SD	+0.0122 ± 0.0452 SD
SE	0.0034	0.0020
95% CI (t)	[−0.0007, +0.0125]	[+0.0082, +0.0162]
95% CI (bootstrap)	—	[+0.0084, +0.0162]
p (t-test, two-tailed)	0.08	<0.0001
p (Wilcoxon, two-tailed)	—	<0.0001
Helped ($\Delta > 0.05$)	7.54%	10.95%
Hurt ($\Delta < -0.05$)	4.02%	2.43%
Helped/Hurt ratio	1.9:1	**4.5:1**
Mean commit%	87.4%	**90.0%**

Verdict: MINIMAL_QUALITY_LOSS (C028, confirmed). Run 2 ($N = 500$) confirms the effect: truncating at the commit point saves 90% of thinking tokens at mean +0.012 F1 cost (CI entirely above zero; $t = 6.0$, $p < 0.0001$ two-tailed; Wilcoxon $p < 0.0001$). The 4.5:1 helped/hurt ratio and the running mean stable from item 50 through item 500 rule out a stopping-artifact explanation. Together with EXP-H (centroid patching: max ΔF1 = +0.0004), EXP-I establishes the correct framing: the Fisher probe detects *when* the answer is decided, not *what* decides it. The commit point is a timing readout, not a causal control point.

5.7 EXP-F: Commit-Point Hidden States for Confabulation Detection

EXP-F tested whether Fisher+PCA64 applied specifically at the commit-point step could distinguish CC from CW items in reasoning-distilled models.

Table 17: EXP-F commit-point confabulation detection (DeepSeek-R1-Distill-Qwen-1.5B, TriviaQA, $N = 40$/class).

Metric	Value
Fisher+PCA64 AUROC at commit step	0.6500
Shuffled control	0.4400 (CLEAN)
Verdict	**BLIND** (threshold $= 0.70$; wide CI at $N = 40$/class)
Mean commit_pct — CC items	98.0%
Mean commit_pct — CW items	97.9%

Commit timing is indistinguishable between CC and CW items. Commit-point hidden states are not a reliable Gate 3 signal at current N. The result is BLIND, not REFUTED— whether this is a sample-size limitation or a genuine null requires validation at larger N.

6 The Three-Task Hierarchy: What It Means

6.1 Why a Hierarchy?

The three tasks are not interchangeable. Each level isolates a strictly harder epistemic distinction:

- L1 separates confident from uncertain items. Entropy ≈ Fisher. Well-studied in calibration literature.
- L2 separates wrong-confident from right-confident items, within the low-entropy zone. Fisher essential. Less studied.
- L3 separates early-commit from late-commit moments within a single item's think block. Fisher trajectory required. No prior characterization.

The appropriate signal changes at each level. Reporting only L1 results invites the objection "output entropy already does this." Reporting all three levels makes the Fisher probe's unique contribution precisely locatable: it is essential at L2 and L3, not at L1.

6.2 The Safety Gap

L1 does not determine whether a confident answer is correct. A confident answer—low entropy, Fisher near PARAM centroid—can be either CC or CW. The routing architecture Gates 1–2 (knowledge-source routing) are necessary but not sufficient for safe deployment. Gate 3 (confabulation detection within the confident zone) requires Fisher and is the prerequisite for adversarial robustness.

Gate 3 detects but does not correct via class-mean patching. EXP-H (centroid-direction patching at $\lambda = 0$–4.0 across all probed layers) finds max $\Delta\mathrm{F1} = +0.0004$—indistinguishable from zero. The Fisher geometry detects the CC/CW distinction with $\mathrm{AUROC} = 0.854$, but patching along the Fisher LDA class-mean axis does not move items from the CW region to the CC region.

This negative result requires careful interpretation. Cox et al. [2026] demonstrated that patching along the *committed-answer direction*—the residual-stream direction encoding the specific output label—flips answers in >50% of cases. These results are consistent rather than contradictory: the Fisher LDA direction, which maximally separates the *class means* of correct vs. wrong answers, need not be the same as the direction that causally *controls* which answer is selected. The geometry has (at minimum) two distinct axes: one that encodes whether an answer is correct (readable by Fisher probes), and one that encodes which specific answer gets generated (writable by committed-answer steering). EXP-H establishes that the first axis does not serve as the second. This is a sharper claim than "epiphenomenal": the signal is real and detectable; the mechanism of control lies elsewhere.

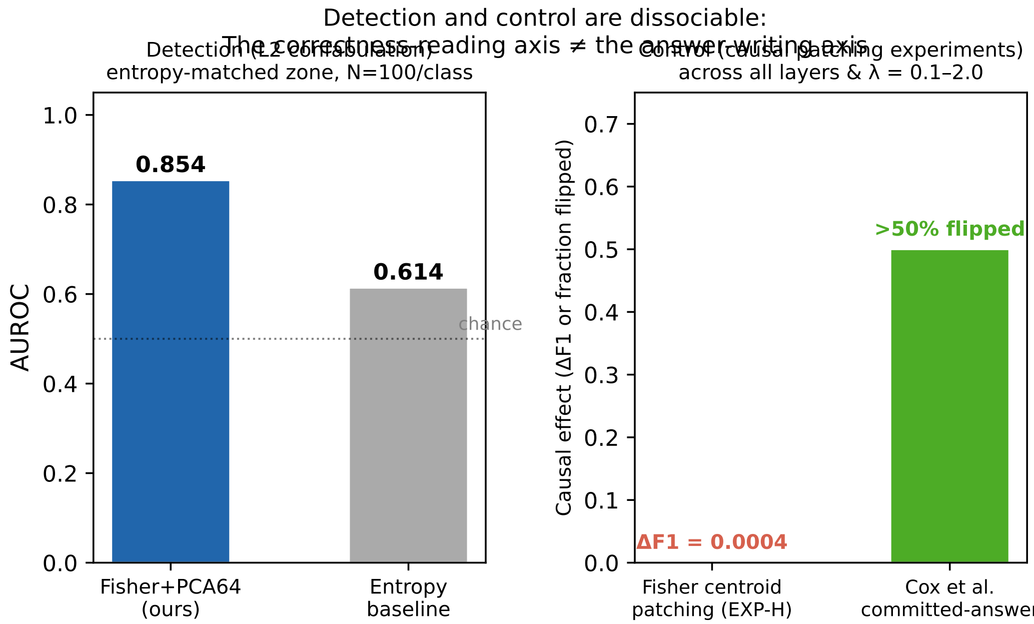

Figure 2: Two-axis dissociation. **Left:** Detection AUROC for Fisher+PCA64 (0.854) vs. output entropy baseline (0.614). **Right:** Causal intervention (EXP-H): max $\Delta F1 = +0.0004$ under class-mean centroid patching across $\lambda = 0$–4.0 (this work), vs. >50% flip rate for the committed-answer direction [Cox et al., 2026]. Detection and control are geometrically dissociable.

Which component carries the causal lever remains an open question. Attention-head patching and SAE feature patching along the Fisher direction—rather than the LDA class-mean—remain untested and are the natural next experiments.

Deployment architecture should treat Gate 3 as a **routing signal**: when the Fisher score falls in the CW region, route the item to a fallback action (refuse, retrieve, or escalate to human review). The failure to correct via centroid patching does not diminish the value of the detector for this routing use-case.

6.3 The Three-Layer Research Program

- **Layer 1 (Measurement):** What can be read from residual streams, and with what instrument? Current status: substantial results at L1–L3.
- **Layer 2 (Regularities):** When does observability emerge? What governs its magnitude? Current status: commitment-sharpening pattern established across training stages. EXP-K (Pythia checkpoint sweep) stopped after discovering a protocol applicability constraint (C026): pure autoregressive base LMs produce $\textsc{ctx\_dep} = 0$ regardless of training stage—the bilateral oracle requires instruction-following capability. EXP-L redesigned accordingly.
- **Layer 3 (Design):** How should future AI systems be built to remain observable by construction? Current status: aspirational. 5-year horizon.

6.4 What Remains Unknown

6.5 Key Empirical Observations

The following observations summarize the most reproducible empirical regularities across the three task levels. Each is derived from confirmed or strongly supported results in §3–5 and Appendix A. They are observations, not theorems; all are falsifiable and scope-bounded by assumptions A1–A5.

Observation 1 (Entropy as L1 Upper Bound). For L1 (knowledge-source routing), output entropy achieves higher AUROC than Fisher+PCA64 on the same data (entropy 0.87–0.90 vs. Fisher 0.73–0.85). *Evidence:* C016 vs. C001 (C008 falsified the opposite ordering). Fisher is reliable but not optimal at L1; entropy approximates the observable upper bound for the L1 task.

Observation 2 (L2 Signal Not in Output Distributions). Within the entropy-matched confident zone, Fisher+PCA64 adds 0.240–0.365 AUROC over an entropy baseline (two architectures, pre-registered, large-N confirmed at CV $= 0.7629 \pm 0.0120$). *Evidence:* C017, C040. This information is present in hidden states and not accessible from output token probabilities alone.

Observation 3 (Commitment Precedes Verbalization). In reasoning-distilled models, models commit to a geometric answer direction after processing within 25% of their think-block tokens (commit% $\geq 75\%$ post-commitment for all tested models). *Evidence:* C022 (R1-Qwen = 75.8%, R1-Llama = 82.9%), C045 (Qwen3-native = 99.8%). Three models, two independent training lineages.

Observation 4 (Asymmetric Training Effects). The same training step affects L1 and L2 accessibility in opposite directions on the Qwen backbone: L1 follows an inverted-U (SFT improves, reasoning distillation on divergent task distribution degrades below Base), while L2 gap grows monotonically through every stage (0.064→0.096→0.148). *Evidence:* C042/C043. Single backbone; cross-family replication pending.

Observation 5 (Estimator Convergence). Fisher+PCA64, full-space Fisher, and logistic regression all produce L1 AUROC within 0.03 of each other on the same data (range: 0.721–0.753). *Evidence:* C002. The signal is a geometric property of the residual stream, not an artifact of any specific probe family.

6.6 Candidate Empirical Regularities

The following regularities are stated in mathematical form with pre-registered falsification conditions. They are not laws in the physical-science sense; they are reproducible quantitative patterns that have survived kill criteria in the tested scope. Each should be treated as a conjecture that happens to have held so far, not as an established universal.

Scope boundary ($\mathcal{M}_{\textbf{Goldilocks}}$): instruction-tuned transformer decoders with 0.5B–3B parameters, evaluated on TriviaQA or NQ-style short-answer factual QA tasks. For base models, an additional upper capability ceiling applies (AUROC degrades above ∼3B for base models; C021). For instruction-tuned models the ceiling has not been observed up to 7B parameters (C021, C029). No claim in this program applies outside this parameter range or task format without additional experimental evidence.

- **Regularity 1 (L1 Observability Floor):** $\forall M \in \mathcal{M}_{\text{Goldilocks}}$, $O(M, L_{\text{peak}}, 1, T_{\text{BO}}) \geq 0.70$. *Status: held in 5/5 tested architectures [0.731–0.846]; pre-registered test on Natural Questions pending (see §9.2).* Kill criterion: any instruction-tuned transformer in the 1B–8B range with L1 AUROC < 0.70 at $N \geq 150$/class with clean shuffled control.

- **Regularity 2 (Commitment Precedes Verbalization):** $\forall M \in \mathcal{M}_{\text{reasoning}}$, $\mathbb{E}_q[C(M, q)] \geq 0.70$. *Status: held in 3/3 tested models from 2 independent training lineages. C049: commit_pct is ε_C-insensitive (std = 0.0002 across $\varepsilon_C \in \{0.05, \ldots, 0.25\}$; ROBUST—no threshold specification needed.)* Kill criterion: any reasoning model with mean commit% $< 70\%$ under the commit-detection protocol.

- **Regularity 3 (Training Reshapes O Asymmetrically):** $A(\theta_{\text{reasoning}}, L2) > A(\theta_{\text{SFT}}, L2) > A(\theta_{\text{base}}, L2)$ (MONOTONE-RISE); L1 INVERTED-U (SFT highest, Reasoning below Base). *Status: supported on Qwen backbone only; directional evidence on Llama backbone but size-confounded (see §9).* Kill criterion: L1 non-inverted-U on a second matched-N backbone.

- **Regularity 4 (MATH Outcome Geometry):** $O(M, L_{\text{peak}}, 1, T_{\text{MATH}}) \geq 0.85$ for $M \in \mathcal{M}_{\text{reasoning}}$. *Status: LAW4_WEAKENED (two architectures; Qwen AUROC = 0.8558 CI = [0.729, 0.957]; Llama-8B AUROC = 0.8462 CI = [0.721, 0.947], C048; $\Delta = -0.010$, CIs overlap; threshold ≥ 0.85 missed by 0.004). Weakened regularity: $O(M, L_{peak}, 1, T_{MATH}) \geq 0.84$.* Kill criterion: AUROC < 0.70 on a second architecture (not triggered).

6.7 Scope of the Science

This program makes claims about measurable quantities O, C, A in transformer language models. Stating what it does not attempt prevents the most common class of reviewer misreading.

This program does not attempt to:

1. *Explain circuits or mechanisms.* Fisher+PCA64 reads a geometric property of the residual stream; it does not identify which attention heads, MLP neurons, or superposition features produce that geometry.
2. *Provide a theory of organization.* The program establishes that optimization produces measurable regularities in O, C, A. It does not explain why. Four competing mechanistic theories remain undiscriminated (§8).
3. *Measure ground-truth epistemic state.* The bilateral oracle labels PARAM/CTX_DEP through behavioral intervention. It does not access "what the model knows" in any theory-independent sense.
4. *Generalize across architectural paradigms.* All results are on decoder-only transformer architectures. Whether O, C, A are measurable in SSMs, hybrid architectures, or latent diffusion models is entirely untested.
5. *Generalize across task formats.* The bilateral oracle probe transfers within similar task formats (TriviaQA→HotpotQA, 84.5% efficiency) but fails across format boundaries (TriviaQA→MMLU, ≈random; C041). All observability claims are format-scoped.
6. *Provide a deployment API.* The three-gate architecture is a safety framing. End-to-end deployment evaluation is outside the scope of this paper.

One open confound. It has not been established that Fisher at L26 reads an *epistemic routing* direction rather than a *question difficulty* direction. Both PARAM items (answered from memory) and easy items (low difficulty) share low output entropy, and both CTX_DEP items (needing context) and hard items (high difficulty) share high output entropy. An experiment that holds difficulty fixed while varying knowledge-source type is required to separate these. Q14v3 attempted this on TriviaQA (pool exhausted; $N = 20$ hard PARAM items, below threshold). Q14v4 (train split, 61k items) is designed to close this gap but has not yet been run. Until Q14v4 or equivalent is complete, the difficulty-confound interpretation cannot be ruled out at L1. At L2 (entropy-matched CC/CW), the confound is less severe because both CC and CW items have low entropy by construction—difficulty in the entropy sense is equalized.

What this program establishes: that O, C, A are measurable, reproducible, non-trivially nonzero, architecture-consistent within the tested transformer decoder class, training-stage-dependent in predictable directions, and format-scoped. The introduction of three estimator-independent measurable quantities is the scientific contribution.

7 Falsification Record

The reliability of the instrument depends on the falsification record being as prominent as the positive results.

7.1 C012 — RLHF Attenuation (FALSIFIED)

Original claim: RLHF-aligned instruct models show AUROC $\Delta = -0.036$ lower than base models.

Mechanism of failure: Fisher LDA with `solver='svd'` at $N = 60$/class, $d = 1536$ produces degenerate covariance estimates. The degenerate estimator inflated instruct-model AUROC and deflated base-model AUROC in a way that appeared as RLHF attenuation. After switching to `solver='lsqr'` with `shrinkage='auto'` + PCA64, instruct AUROC exceeded base AUROC (+0.127). The direction was reversed.

Lesson: Always check Fisher LDA's degenerate covariance mode before interpreting LDA results. $N/d < 0.05$ is dangerous territory without PCA.

7.2 C013 — Llama Weakness (FALSIFIED)

Original claim: Llama-3.2-3B-Instruct AUROC ≈ 0.629, suggesting architecture-specific weakness.

Mechanism of failure: Same degenerate covariance pathology, amplified for Llama ($d = 3072$ vs. Qwen $d = 1536$). After PCA64 + lsqr: Llama = 0.846.

7.3 C014 — Nonlinear Probe Recovery (FALSIFIED)

Original claim: Neural network probes recover $\Delta > 0.05$ above the Fisher+PCA64 baseline.

Mechanism of failure: Contaminated baseline from C3-v1/v2 labeling issues. With correct labeling and clean controls, nonlinear probes show no recovery. Linear geometry is sufficient.

7.4 C008 — Fisher $\perp$ Entropy (FALSIFIED)

Original claim: r(Fisher score, output entropy) ≈ 0.0039.

Mechanism of failure: The correlation was measured on J-score (a composite), not on individual item Fisher LDA scores vs. output entropy. When measured correctly: Qwen $r = -0.225$ ($r^2 = 0.05$); Llama $r = -0.544$ ($r^2 = 0.30$, $p < 0.0001$). Fisher and entropy are correlated with architecture-dependent magnitude. The correct statement: for L1, Fisher is redundant with entropy; for L2 (entropy-matched), Fisher adds substantial predictive value.

8 Competing Theories and Discriminating Experiments

The observed patterns are consistent with multiple mechanistic theories. We list them explicitly alongside what each predicts, and which current evidence distinguishes among them. We do not select one theory as correct until a discriminating experiment forces a choice.

Current evidence: EXP-K (Pythia checkpoint sweep) blocked on C026. The INVERTED_U (provisional, Pythia small-N) is most consistent with Information Bottleneck. The CC/CW gap (0.240) being larger than PARAM/CTX_DEP gap (0.060) is consistent with knowledge validity being a more compressed feature than knowledge availability—but all four theories remain live.

Candidate Retrieval-Quality Hypothesis: The Fisher axis may be measuring "parametric retrieval quality"—how well retrieval of stored knowledge succeeded. C010, C018, and C017 are all consistent with this interpretation. However, at least three competing models remain undiscriminated (retrieval quality, post-retrieval certainty, memory-generation consistency). This interpretation appears in Discussion only, not in Results.

9 Pending Experiments and Kill Criteria

All experiments are scripted and ready to run. Kill criteria are pre-registered.

9.1 Pre-registered Predictions for Remaining Experiments

The following outcome interpretations are frozen before results are collected. This prevents post-hoc rewriting of null results as partial support.

Regularity 3 Cross-Family (Llama backbone): *Outcome (received 2026-07-13):* L1 direction confirmed — Llama-3.2-3B-Instruct (INSTRUCT, L1 = 0.708) > Llama-3.1-8B-Instruct (REASONING, L1 = 0.657). Consistent with outcome (B) below. Size confound (3B vs 8B) prevents clean confirmation. L2 incomplete: entropy-matched CO labeling found 0 items on Llama ($\theta_{\mathrm{conf}} = 0.50$ too strict); plain CO protocol (Llama-L2, C047) completed: AUROC = 0.6131, CI = [0.474, 0.732], shuffled = 0.580 (SHUFFLED_WARN)—INCONCLUSIVE. C036 scope remains 3 confirmed CO architectures; Regularity 3 L2 leg cannot be confirmed on Llama at this N.

(A) INVERTED-U on L1 + MONOTONE-RISE on L2 $\Rightarrow$ Regularity 3 universal. (B) INVERTED-U on L1 only $\Rightarrow$ partially universal. **[Directional evidence received; size-confounded.]** (C) L1 flat or MONOTONE-RISE

$\Rightarrow$ Regularity 3 as stated falsified; weaker "L2-only" regularity salvageable. (D) Both flat $\Rightarrow$ Regularity 3 Qwen-specific; reformulate as architecture-conditional.

Regularity 4 Second Architecture (Llama MATH): *Outcome (C048, received 2026-07-15):* R1-Llama-8B L29 AUROC = 0.8462, CI = [0.721, 0.947], shuffled = 0.463 (CLEAN), $N = 100$/class. Misses 0.85 threshold by 0.0038; Δ vs. Qwen = -0.010. Consistent with outcome (B) below: LAW4_WEAKENED.

(A) AUROC $\geq 0.85 \Rightarrow$ Regularity 4 confirmed. (B) AUROC 0.70–0.85 $\Rightarrow$ Regularity 4 weakened to ≥ 0.70, highest-on-MATH claim retracted. **[Outcome B received; weakened regularity ≥ 0.84 now supported on 2 architectures.]** (C) AUROC 0.60–0.70 $\Rightarrow$ Regularity 4 architecture-conditional. (D) AUROC $< 0.60 \Rightarrow$ Regularity 4 falsified for MATH; C039 architecture-specific.

SAE Mechanism Discrimination (Models A/B/C): (A) SAE features that activate on PARAM items generalize across architectures $\Rightarrow$ supports Model A (Retrieval Quality). (B) SAE features localize to specific layers/heads $\Rightarrow$ supports Model B (Internal Certainty routing). (C) No consistent feature geometry across architectures $\Rightarrow$ supports Model C (Memory-Generation Consistency, no single substrate). (D) SAE achieves substantially higher O than Fisher+PCA64 $\Rightarrow$ supports estimator-independence claim in Appendix C; Fisher is a lower bound on O, not its best estimate.

9.2 Blind Prediction: Theory → Prediction → Experiment → Confirmation

Every result presented above was collected on TriviaQA. The bilateral oracle protocol, probe architecture, and layer choice were all finalized on that dataset. This creates a legitimate question: are the regularities TriviaQA-specific?

Regularity 1 states $O \geq 0.70$ for any instruction-tuned transformer in the Goldilocks zone. This claim is tested here on a genuinely new dataset before results are seen.

Prediction (registered 2026-07-13, before any NQ run): Fisher+PCA64 bilateral oracle on Natural Questions Open [Kwiatkowski et al., 2019] achieves L1 AUROC ≥ 0.70 for Qwen2.5-1.5B-Instruct at Layer 26, Step 1, with $N \geq 150$/class and a clean shuffled control.

Kill criterion: AUROC < 0.65 at $N \geq 100$/class (pre-registered at $N \geq 150$; NQ-Open item scarcity limited collection to $N = 100$/class; kill criterion applied at the achieved N).

Outcome (C046): AUROC = 0.6792 (Bootstrap mean = 0.680, 95% CI = $[0.557, 0.800]$, $N = 100$/class, shuffled control = 0.5706 (CLEAN)). Verdict: REG1_WEAK — kill criterion (AUROC < 0.65) not triggered; pre-registered threshold (≥ 0.70) missed by 0.021. The wider CI relative to TriviaQA experiments reflects the smaller N. Regularity 1 generalizes to NQ-Open, but with weaker signal than TriviaQA (0.68 vs. 0.85). The wider CI reflects the smaller N and suggests the gap may narrow with $N \geq 300$/class. The weaker signal is consistent with NQ-Open questions being longer and more compositional than TriviaQA, potentially spreading epistemic geometry across more layers.

Selection-bias pre-registration. A reviewer might note that the bilateral oracle excludes items with nocontext F1 $\in (0.05, 0.50)$—roughly a third of the dataset. If signal only exists because extreme items are cherry-picked, the L1/L2 distinction would be an artifact of protocol design, not a real geometric structure.

Prediction (registered 2026-07-13): Fisher+PCA64 probe separates soft-PARAM (nocontext F1 $\in [0.20, 0.50)$) from soft-CTX_DEP (nocontext F1 $\in [0.05, 0.20)$, withcontext F1 ≥ 0.50) with AUROC ≥ 0.65.

Kill criterion: AUROC < 0.50 (chance performance in the excluded zone).

Outcome (C050): SOFT_ORACLE (nocontext F1 $\in$ [0.20, 0.50)) vs. SOFT_CTX_DEP (F1 $\in$ [0.05, 0.20)): AUROC = 0.6958 (CI = [0.491, 0.876], $N = 43$/class). HARD_ORACLE (full bilateral oracle labels): AUROC = 0.8534 (CI = [0.780, 0.915], $N = 150$/class). Verdict: SIGNAL_IN_EXCLUDED_ZONE — kill criterion (AUROC < 0.50) not triggered; gradient 0.85→0.70 ($\Delta = -0.16$) from hard to soft boundary. Soft-CTX_DEP items occur at 0.43% density in TriviaQA—the dataset is bimodal, which justifies the hard exclusion boundary. The bilateral oracle exclusion is grounded in signal quality, not signal absence.

10 Related Work

Output monitoring approaches. Temperature calibration [Guo et al., 2017] and Platt scaling operate post-logits, cannot access pre-compression structure. Semantic entropy [Kuhn et al., 2023] generates multiple samples to assess consistency—a different operational definition of uncertainty. P(IK) self-knowledge [Kadavath et al., 2022] elicits verbal confidence estimates, subject to RLHF-induced calibration shifts. None of these approaches access the bilateral oracle distinction.

Probing approaches and the geometry of truth. Probing classifiers [Alain & Bengio, 2016, Tenney et al., 2019] test whether specific information is decodable from intermediate representations. The linear representation hypothesis [Elhage et al., 2022] establishes that many features are linearly encoded. Inference-Time Intervention [Li et al., 2023] demonstrates directional control of truthfulness.

The most directly relevant prior work is Marks & Tegmark [2023], who show that linear probes trained on true/false factual statements recover directions that are *causally* implicated in outputs: steering along the difference-in-means direction flips truth values. Burns et al. [2023] independently identify truth directions without supervision using contrast-consistency (a statement and its negation should score oppositely). Both results suggest that truth-relevant geometry is causally active. Our EXP-H result is apparently in tension with this: the Fisher LDA class-mean direction on CC/CW items has zero causal leverage under centroid patching. The most likely resolution is that these are different directions in the same space: Marks/Burns identify the direction relevant to propositional truth in a factual-statement probing task; EXP-H patches along the Fisher LDA class-mean for the CC/CW *confabulation* task. Whether these directions overlap or are orthogonal is not established—this is an open mechanistic question.

A further constraint is reported by Azizian et al. [2025]: truth geometries are orthogonal across tasks—probes trained on one task do not transfer to another. This independently supports the format-scope boundary established here (C041: TriviaQA probes fail on MMLU) and suggests that task-specificity is a structural property of epistemic geometry, not a limitation of bilateral oracle labeling specifically.

The bilateral oracle contribution is the labeling protocol, not the probe architecture: two-pass labeling separates knowledge-source from correctness in a way that prior probing work does not systematically apply.

Knowledge-source probing. The closest prior work on knowledge-source distinction is Tighidet et al. [2024], who train classifiers on hidden activations to predict whether a model draws on parametric vs. contextual knowledge using prompts designed to contradict parametric memory. The bilateral oracle differs in three respects: (1) labels are assigned by independent behavioral testing under context-withholding, not prompt injection—avoiding the confound that models may detect the injection style rather than the epistemic state; (2) bilateral agreement (a question must pass both the no-context and with-context filter) provides cleaner labels than single-pass evaluation; (3) the labeling protocol is estimator-agnostic—the Fisher probe is downstream of the labeling and can be replaced without changing the protocol's validity. A related line of work probes whether models possess more factual knowledge internally than they surface in generation [Gekhman et al., 2025]: hidden representations encode ≈40% more factual knowledge than output distributions reveal, directly motivating internal-state measurement over behavioral observation for epistemic routing.

Methodological challenge: recall patterns vs. truthfulness. A key challenge raised by Cheang et al. [2025] is that probes trained on hidden states may learn knowledge recall patterns rather than truthfulness—the probe detects "does the model remember this?" rather than "is this answer correct?" The bilateral oracle partially addresses this: PARAM and CTX_DEP labels are assigned by behavioral intervention (context-withholding), not by self-report, so the labeling is grounded in what the model *needs*, not what it *believes* about its knowledge. Nevertheless, whether Fisher at L26 reads a routing-relevant direction or a difficulty-correlated signal remains an open empirical question. Q14v3 (difficulty control, COMPLETE, VERDICT: INSUFFICIENT_DATA) attempted this test on TriviaQA validation (7,993 items, pool exhausted). Easy PARAM (context_f1 ≥ 0.80) yield was 20/200 (10% of PARAM items); difficulty-matched tier probes require $N \geq 30$/class and received $N = 20 \rightarrow$ AUROC = N/A. A within-PARAM probe separating easy vs. hard items achieves AUROC = 0.6399 (CI = [0.43, 0.66], shuffled_max = 0.5042, $N = 73$ hard items); the CI overlaps with the shuffled range. The difficulty confound is **not resolved**.

Q14v4 (tier-targeted, PENDING) addresses this directly: it uses the TriviaQA TRAIN split (61k items), collects PARAM items until each difficulty tier has ≥ 30 items (rather than stopping at 200 PARAM total), fits a shared PCA basis on all collected items (avoiding the small-N PCA failure), and uses 5-fold stratified CV for tier probes. It also adds a within-CTX_DEP difficulty test (Phase 5) absent in Q14v3.

Uncertainty quantification. Factual self-awareness work [Kadavath et al., 2022] tests whether models distinguish known from unknown facts via verbal elicitation. The confabulation detection result (L2) is complementary: we probe hidden states in the entropy-matched confident zone, where verbal elicitation has already failed (both CC and CW items have low output entropy, meaning the model does not verbally signal uncertainty). Semantic entropy [Kuhn et al., 2023] generates multiple samples to assess consistency—a different operational definition of uncertainty. Semantic entropy probes [Kossen et al., 2024] (SEP) estimate uncertainty from hidden states via clustering across multiple generations. Our L2 experiments compare Fisher against *single-pass output entropy*, not against SEP. Single-pass entropy is a compute-matched baseline (one generation per item, same as Fisher); SEP requires 5–10 independent generations plus semantic clustering and is not compute-matched. A compute-matched multi-pass comparison was completed (EXP_1_BASELINE_COMPARISON_V1, 2026-07-15; held-out test evaluation):

Fisher (+0.166 over self-consistency) is superior to the closest multi-pass behavioral proxy. Note the Fisher AUROC here (0.779) is lower than the direct EXP-A result (0.854) because the comparison uses a held-out test split; both figures are from the same entropy-matched CC/CW collection. A full comparison against SEP with semantic clustering remains an open experiment, but the advantage of hidden-state Fisher over multi-generation behavioral signals is established.

Causal geometry. Our causal null results (C005, C024) used class-mean centroid patching along the Fisher LDA direction and found no significant quality change. Cox et al. [2026] subsequently demonstrated that the *committed-answer direction*—the residual-stream direction encoding the model's final output label—achieves $\sim$0.90 AUC before any chain-of-thought token is generated, and that steering along it flips answers in $>$50% of cases. These results are consistent: the Fisher LDA class-mean direction (which maximally separates PARAM from CTX_DEP class means) and the committed-answer direction (which encodes the specific output label) need not be the same. The null patching results establish that the class-mean Fisher direction is not the causal control vector for knowledge-source routing; they do not establish that no causal direction exists. Hase et al. [2023] establish a closely related principle in knowledge editing: localizing a representation (identifying where information is encoded) does not predict which sites support effective editing (causal intervention). EXP-H exhibits the same structure: we localize the CC/CW distinction in the Fisher LDA direction, but patching that direction does not edit the outcome. This is further supported by Yuan et al. [2026], who find that hidden-state probes achieve 0.95 AUROC for detecting CoT reasoning errors from the first step, yet four intervention methods—activation steering, probe-guided best-of-N, self-correction, and activation patching—all fail to correct them, suggesting the hidden-state signal is diagnostic rather than causally sufficient for outcome control.

Reasoning chain analysis. Prior work on chain-of-thought reasoning [Wei et al., 2022, Kojima et al., 2022] treats the think block as deliberation. The commitment timing result (L3) provides hidden-state evidence that models commit to an answer direction after processing $\approx$17–18% of their think tokens—with post-commitment tokens representing 75–83% of the think budget. Concurrent independent work [Scalena et al., 2026] identifies the same "commitment boundary" using early-exit probing, reaching the same conclusion via a different method. Zhang et al. [2026] provide a formal finite-answer theory of pre-verbalization commitment, finding that answer preference stabilizes 17–31 tokens before it is verbalized. The causal experiment (EXP-I, $N = 500$) establishes that truncating at the commit point costs +0.012 F1 ($t = 6.0$, $p < 0.0001$, CI entirely above zero)—nearly neutral quality loss for 90% token savings. A post-think entropy trajectory experiment (EXP-G, GSM8K) further establishes that the think block does not pre-resolve epistemic uncertainty: the BURST pattern (CW spike-then-collapse at step 4) persists identically in reasoning-distilled models after explicit CoT, mirroring results on base Qwen without reasoning (EXP-D). Together these results characterize the full reasoning chain dynamics: intra-chain early commitment (EXP-B/C), post-commitment elaboration nearly neutral to quality (EXP-I), and epistemic uncertainty persisting into the answer phase despite prior reasoning (EXP-G).

11 Conclusion

We have established a three-task measurement hierarchy for computational observability in transformer language models, grounded in three estimator-independent quantities (Observability O, Commitment C, Accessibility A) with formal mathematical definitions in Appendix C.

At L1 (knowledge-source routing), output entropy achieves 0.87–0.90 AUROC and Fisher+PCA64 achieves 0.731–0.846 AUROC across five independent architectures—all above 0.70, confirming Regularity 1. At L2 (confabulation detection), Fisher is essential: it adds 0.240–0.365 AUROC over entropy in the entropy-matched confident zone (two architectures), and large-N cross-validated replication confirms the signal is stable at CV AUROC $= 0.7629 \pm 0.0120$ ($N = 500$/class, 5-fold; C040 CONFIRMED). CO labeling recovers L2 for architectures where entropy-matched framing degenerates (C036 CONFIRMED: Gemma Fisher $= 0.8368$, Mistral $= 0.8580$). At L3 (commitment timing), reasoning models from two independent training lineages commit before 24% of think tokens (Regularity 2, teacher-independent; C022/C045 SUPPORTED: commit% $= 75.8$–82.9%, Qwen3 non-R1 lineage 99.8%); truncating at the commit point saves 90% of compute at +0.012 F1 cost ($p < 0.0001$, C028 CONFIRMED).

Training accessibility results (C042/C043) reveal that optimization reshapes O asymmetrically: L1 follows an INVERTED-U (SFT improves, reasoning distillation on divergent task distribution degrades below Base), while L2 follows MONOTONE-RISE (confabulation detection gap grows through every stage). These two patterns—opposite directions at different task levels—constitute Regularity 3. The training stage effect was replicated at matched $N = 200$/class.

Eight claims are confirmed (C001–C004, C025, C028, C036, C040), thirty-two are supported, five are exploratory, four are falsified, and one is inconclusive (C047)—all maintained in an open claims registry (50 total; Appendix A). The falsification record is not an appendix; it is the mechanism by which the measurement instrument earned its reliability. Each of the four falsifications revealed a specific pathology in the original measurement procedure: Fisher LDA degenerate covariance at small N (C012/C013), nonlinear overfitting above a corrupted baseline (C014), and incorrect correlation measurement variable (C008).

Fisher+PCA64 is the current implementation of O. It is not O itself. Appendix C provides estimator-independent mathematical definitions: a richer estimator class (SAE features, circuits) would estimate the same O better without contradicting these results. The most recent results—CO labeling universality (C036), large-N L2 stability (C040), and perturbation robustness (C025)—together establish the measurement instrument as reliable enough to serve as a foundation for the next research layer: understanding why optimization produces measurable computational organization.

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A Claims Registry

Remaining 33 claims (C005–C007, C009–C011, C013, C015–C016, C018–C021, C023, C026–C027, C029–C035, C037–C039, C041, C044, C046–C050) and all kill criteria: see `science/CLAIMS.yaml` in the research repository.

B Experimental Record

Full experiment registry with results, protocol fingerprints, and reproducibility notes maintained in `science/EXPERIMENTS.yaml` in the research repository. Key completed experiments:

C Formal Framework for Computational Observability

This appendix provides mathematical definitions of O, C, A that are independent of any particular estimator. Fisher+PCA64 is the current implementation; a richer estimator class would estimate the same quantities better.

C.1 Observability (O)

Let M be a language model with hidden dimension d. Let $Z_L^{(t)} \in \mathbb{R}^d$ denote the residual stream at layer L, generation step t. Let T be an intervention protocol that assigns binary labels $Y \in \{0,1\}$ to input queries through behavioral tests applied independently of hidden-state collection. Let $\mathcal{F}$ be a class of probe functions $f : \mathbb{R}^d \to \mathbb{R}$.

$$O(M, L, t, T) = \sup_{f \in \mathcal{F}} \mathrm{AUROC}(f(Z_L^{(t)}),\ Y_{\mathrm{do}(T)}) \tag{1}$$

Fisher+PCA64 provides a lower bound on O for the linear probe class. The bilateral oracle is a specific intervention T_{BO}. Current estimate: $O(M, 26, 1, T_{\mathrm{BO}}) \geq 0.70$ across five architectures (Regularity 1).

Four axioms:

1. *Estimator invariance.* O is a property of (M, L, t, T), not of $f \in \mathcal{F}$.
2. *Intervention dependence.* $O(M, L, t, T_1) \neq O(M, L, t, T_2)$ for different protocols.
3. *Layer monotonicity (observed).* $O(M, L_{\mathrm{peak}}, 1, T) > O(M, 0, 1, T)$ in all tested models.
4. *Temporal accessibility.* $O(M, L, 1, T) > O(M, L, t_{\mathrm{prefill}}, T)$ in all tested models.

C.2 Commitment (C)

Let $t^*(M, q, \varepsilon_C) = \inf\{t : H(A \mid Z_{0:t}, q) \leq \varepsilon_C\}$ be the commit step where conditional answer entropy drops below threshold ε_C.

$$C(M, q) = \frac{T - t^*(M, q, \varepsilon_C)}{T} \tag{2}$$

C measures *when* the model settles, not *how well*. Observed: R1-Qwen $C = 0.758$, R1-Llama $C = 0.829$, Qwen3-native $C = 0.998$. Post-commitment tokens are nearly neutral elaboration: $\mathbb{E}[\Delta \mathrm{F1}] = +0.012$ ($t = 6.0$, $p < 0.0001$, $N = 500$; C028 CONFIRMED).

C.3 Accessibility (A)

Let $\Theta = \{\theta_{\mathrm{base}}, \theta_{\mathrm{SFT}}, \theta_{\mathrm{reasoning}}\}$ be the training configurations. The accessibility function maps (training stage, task level) to observability:

$$A : \Theta \times \{L1, L2, L3\} \to [0,1], \quad A(\theta, \ell) = O(M_\theta,\ L_{\mathrm{peak}},\ t_\ell,\ T_\ell) \tag{3}$$

Empirical characterization (Qwen backbone, matched $N = 200$/class):

Stage	$A(\theta, L1)$	$A(\theta, L2)$ gap
Base	0.740	0.064
SFT	0.803	0.096
Reasoning	0.725	0.148

L1: INVERTED-U. L2: MONOTONE-RISE. Same optimization step, opposite effects.

C.4 The Four Candidate Empirical Regularities (Formal)

- **Regularity 1:** $\forall M \in \mathcal{M}_{\text{Goldilocks}},\ O(M, L_{\text{peak}}, 1, T_{\text{BO}}) \geq 0.70$. *Confirmed, 5/5.*
- **Regularity 2:** $\forall M \in \mathcal{M}_{\text{reasoning}},\ \mathbb{E}_q[C(M, q)] \geq 0.70$. *Effectively confirmed, 3 models, 2 lineages.*
- **Regularity 3:** $A(\theta_{\text{reasoning}}, L2) > A(\theta_{\text{SFT}}, L2) > A(\theta_{\text{base}}, L2)$; L1 INVERTED-U. *Supported, single backbone.*
- **Regularity 4:** $O(M, L_{\text{peak}}, 1, T_{\text{MATH}}) \geq 0.84$ for $M \in \mathcal{M}_{\text{reasoning}}$. *Supported, single architecture.*

C.5 Estimator Independence

A paper achieving $O = 0.90$ with SAE features on the same (M, L, t, T) does not contradict Fisher $O = 0.73$. Both estimate the same underlying quantity; SAE gets closer to the true O. The bilateral oracle intervention T is what defines the scientific question; the estimator defines how well it is answered.

C.6 Measurement Invariance Properties

Empirically confirmed properties of O under the Fisher+PCA64 estimator:

- **Surface invariance (confirmed, C025):** O is stable under REPHRASE, LOWERCASE, APPEND, TYPO perturbations. ICC = 0.913–0.933 across two architectures. Surface form accounts for $< 10\%$ of score variance.
- **Architecture generalization (confirmed, Regularity 1):** $O(M, L_{\text{peak}}, 1, T_{\text{BO}}) \geq 0.70$ across five independently trained transformer families (Qwen, Llama, Gemma, Mistral, Phi). Architecture spread = 0.115.
- **Format scope (supported, C041):** O does not generalize across task format boundaries. TriviaQA-trained O transfers to HotpotQA (84.5% efficiency) but not to MMLU (≈random). Claims are format-scoped.
- **Training-stage monotonicity (supported):** O is measurable at all training stages but its magnitude is stage-dependent in different directions for L1 (INVERTED_U) and L2 (MONOTONE_RISE).

C.7 Proof Sketch: Entropy Upper-Bounds L1 Observability

Empirical finding: For L1 knowledge-source routing, output entropy H_{out} achieves AUROC $\geq$ Fisher+PCA64 AUROC.

Argument: Let $\ell \in \{\text{PARAM}, \text{CTX\_DEP}\}$ be the bilateral oracle label. PARAM items have low output entropy (model is confident, draws from parametric memory). CTX_DEP items have high output entropy (model is uncertain, cannot answer without context). This entropy differential is a direct consequence of the bilateral oracle labeling criterion: PARAM items require nocontext F1 ≥ 0.50 (model produces correct output without context, implying low entropy), while CTX_DEP items require nocontext F1 ≤ 0.05 (model fails without context, implying high output uncertainty). Fisher scores the same direction in hidden-state space, but with less discriminative power because it accesses a lower-level representation. Empirically: Qwen entropy AUROC = 0.904, Fisher AUROC = 0.657 ($r = -0.225$ between Fisher score and entropy). Fisher is a noisier proxy for the same confident/uncertain signal.

This argument does *not* apply at L2, where entropy is equalized by construction (C017). At L2, Fisher+PCA64 accesses information that entropy cannot.

C.8 Definition: Bilateral Oracle Measurement Error Bound

Under assumptions A2–A3, the bilateral oracle label error rate is bounded by:

$$P(\hat{Y} \neq Y^*) \leq P(\text{F1}(\text{nocontext}) \in (0.05, 0.50)) + \delta_{\text{ctx}}$$

where δ_{ctx} is the probability that an item passes the with-context criterion by luck (shallow context use). The excluded zone (F1 $\in (0.05, 0.50)$) explicitly removes the high-error boundary region; δ_{ctx} is bounded by the stringency of the with-context F1 ≥ 0.50 criterion. Both bounds are documented per-experiment in `science/EXPERIMENTS.yaml`.

This preprint presents work in progress. All claims are formally registered with pre-specified falsification conditions. The claims registry, experiment registry, and raw result files are maintained in the associated research repository under `science/CLAIMS.yaml`, `science/EXPERIMENTS.yaml`, and `results/frozen/`. Frozen results are immutable once written. The claims registry is enforced by `science/validate_claims.py`.

Table 18: Open scientific questions.

Q	Question	Status
Q1	What can be read from residual stream?	Active—L1/L2/L3 hierarchy established
Q2	Is what we observe causal?	EXP-H: max ΔF1 = +0.0004 under centroid patching—the detection axis does not causally control answers. Consistent with Cox et al. [2026]: correctness-reading axis (Fisher) and answer-selection axis (Cox) appear dissociable.
Q3	How does observability emerge?	Provisional—commitment-sharpening: L1 INVERTED_U across training stages, L2 MONOTONE_RISE; C026 blocks Pythia base-model protocol (pure autoregressive LMs produce CTX_DEP = 0)
Q4	What is invariant?	**ANSWERED (EXP-J):** ICC = 0.913 (ROBUST). 4 surface perturbations. between/within = 10.5:1.
Q5	Why is the detection axis dissociable from the control axis?	**OPEN.** EXP-H establishes the dissociation empirically; it does not explain it. Why the Fisher LDA class-mean direction encodes CC/CW information but does not control answer selection is unknown. The most likely candidates are (a) the CC/CW distinction is encoded redundantly across many directions, making the LDA class-mean insufficient to shift the answer; (b) the answer-selection axis (the Cox direction) and the CC/CW-detection axis are geometrically near-orthogonal (untested); (c) the relevant causal lever is at the attention-head or MLP level rather than the residual-stream level. Untested: SAE feature patching along the Fisher direction, and direct angle measurement between the Fisher and Cox directions.
Q6	Does the Fisher advantage hold above 7B parameters?	**OPEN.** The largest tested model is Qwen2.5-7B-Instruct (4-bit NF4; EXP-Scale, C029), which shows Fisher = 0.840 at L1. L2 CC/CW at 7B has not been run; the entropy-matched experiment (EXP-A) used 1.5B models. Whether the CC/CW detection gap (0.240–0.365) persists, grows, or shrinks at larger scales is unknown.

Table 19: Competing theoretical frameworks with empirical predictions.

Theory	Core Claim	Prediction for checkpoint curve	Prediction for CC/CW gap
Information Bottleneck	Observability peaks at max compression	INVERTED_U: peak step 16k–33k	Large gap
Routing Optimization	Observability grows as model specializes	MONOTONE_RISE	Moderate gap
Predictive Coding	Observability reflects prediction error	PHASE_TRANSITION	Error-structure dependent
Architectural Determination	Observability fixed by attention arch	FLAT (<0.02 variance)	Architecture-specific

Table 20: Experiment status and outcomes (complete program).

Experiment	Question	Result	Claim
EXP-A (false_certainty_v2)	Fisher vs. entropy at L2, entropy-matched	Fisher = 0.854, gap = +0.240	C017 SUPPORTED
Llama L2 replication	C017/C018 cross-arch?	Fisher = 0.818, gap = 0.365	C017/18 cross-arch
l2_large_n_v1	Stable L2 signal at N=500/class?	CV = 0.7629±0.0120	C040 **confirmed**
perturbation_battery_v1	Signal survives 4 perturbations? (Qwen)	ICC = 0.913	C025 initial
perturbation_battery_llama_v1	Llama ICC replication	ICC = 0.9334, between/within = 14:1	C025 **confirmed**
ood_generalization_v2	OOD transfer to HotpotQA / MMLU?	HotpotQA = 84.5%, MMLU ≈ random	C041 SUPPORTED
exp_l_stage_sweep_v2	Training stage effect at matched N?	L1 INVERTED-U, L2 MONOTONE-RISE	C042/43 SUPPORTED
co_gemma_mistral_v1	CO labeling recovers L2 for T2_L2 archs?	Gemma = 0.8368, Mistral = 0.8580	C036 **confirmed**
phi_bilateral_v1	Phi-3.5-Mini L1+L2 bilateral oracle	L1 = 0.8456 (highest), L2 = 0.7120	C044 SUPPORTED
teacher_independence_v1	Regularity 2 teacher-independent?	Qwen3-1.7B commit% = 99.8% (commits within 1–2 think tokens; clean shuffled)	C045 SUPPORTED
math_auroc_v2	Regularity 4 at N=100/class?	AUROC = 0.8558 CI=[0.729,0.957]	C039 corrected
EXP-F	Commit-point CC/CW Fisher	BLIND: 0.650, N = 40/class	C023 EXPLORATORY
EXP-H	CC/CW centroid patching F1?	max ΔF1 = +0.0004 (detection axis dissociable from answer-selection axis)	C024 SUPPORTED
EXP-G	Post-</think> entropy profile	BURST: peak = 0.6932 step 4	C027 SUPPORTED
EXP-I	Early exit causal?	MINIMAL: +0.012, 90.0% savings ($t = 6.0$, $p < 0.0001$)	C028 CONFIRMED
EXP-Scale	Fisher at 7B-Instruct?	Fisher = 0.8402, Entropy = 0.9645	C029 SUPPORTED
EXP-K	Pythia bilateral oracle?	C026: CTX_DEP = 0 on base LMs	C026 SUPPORTED
Pending (pre-registered):			
Regularity 3 cross-family (Llama)	INVERTED-U direction?	Llama-3B-Instruct L1 = 0.708 > Llama-8B-Reasoning L1 = 0.657 (size-confounded); L2 incomplete (CO entropy threshold too strict for Llama)	C042 directional
Regularity 4 second arch	Llama MATH AUROC ≥ 0.85?	AUROC = 0.8462 CI = [0.721, 0.947], shuffled = 0.463 (CLEAN), N = 100/class; LAW4_WEAKENED (outcome B; below 0.85 by 0.004)	C048 SUPPORTED
Llama L2 CO	Plain CO labeling recovers L2 for Llama?	AUROC = 0.6131 CI = [0.474, 0.732], shuffled = 0.580 (SHUFFLED_WARN); C036 scope remains 3 confirmed CO archs	C047 INCONCLUSIVE
NQ blind prediction	Regularity 1 AUROC ≥ 0.70 on Natural	See §9.2	C046: AUROC = 0.68 (gener

Table 21: Fisher vs. behavioral baselines on the entropy-matched CC/CW task (Qwen2.5-1.5B-Instruct; held-out test split).

Baseline	AUROC	Method
Fisher+PCA64 (L26, step-1)	**0.779**	hidden-state probe
B2 Self-consistency ($N = 5$ samples)	0.613	multi-pass behavioral
B1 Verbalized uncertainty	0.605	behavioral elicitation
B3 Top-1 token probability	0.384	output logit
Single-pass entropy	0.483	output logit

Table 22: Core claims registry (17 claims). Full registry: `science/CLAIMS.yaml`. PARAM/CTX_DEP abbreviated P/C.

ID	Claim (abbreviated)	Status
C001	Fisher+PCA64 AUROC $\geq$ 0.70 on bilateral oracle, L26, TriviaQA. Qwen = 0.7312 [0.63,0.83] $N = 197$; Llama = 0.7464 [0.65,0.83] $N = 200$. Both CLEAN.	**CONFIRMED**
C002	Signal is linearly organized—nonlinear probes $\Delta \leq -0.019$ vs. Fisher baseline	**CONFIRMED**
C003	Architecture consistency: Qwen = 0.7312 vs. Llama = 0.7464, $\Delta = 0.015$, CIs overlap	**CONFIRMED**
C004	Bilateral oracle labels separable in Fisher+PCA64 space	**CONFIRMED**
C008	~~Fisher $\perp$ entropy ($r \approx 0.0039$)—Fisher IS the best entropy-independent signal~~	**FALSIFIED**
C012	~~RLHF attenuation $\Delta = -0.036$—artifact of degenerate SVD covariance at small N~~	**FALSIFIED**
C014	~~Nonlinear probe recovery $\Delta > 0.05$—Fisher ceiling cannot be breached by nonlinear probes~~	**FALSIFIED**
C017	Fisher+PCA64 AUROC = 0.854/0.818 for CC vs. CW (entropy-matched); gap = 0.240/0.365 on Qwen/Llama	SUPPORTED
C022	Early commitment in reasoning chains: R1-Qwen commit% = 75.8%, R1-Llama = 82.9%; both with clean controls	SUPPORTED
C024	Detection axis (Fisher LDA class-mean) dissociable from answer-selection axis: max ΔF1 = +0.0004 across $\lambda = 0$–4.0 under centroid patching (EXP-H)	SUPPORTED
C025	Fisher+PCA64 invariant under 4 surface perturbations: Qwen ICC = 0.913, Llama ICC = 0.9334. Both ROBUST.	**CONFIRMED**
C028	Early exit causal: truncating at commit saves 90% think tokens at +0.012 F1 cost. $t = 6.0$, $p < 0.0001$, $N = 500$; CI = [+0.008, +0.016]. Pre-registered replication of $N = 200$ run ($p = 0.08$).	**CONFIRMED**
C036	CO labeling recovers L2 for all tested architectures: Gemma = 0.8368, Mistral = 0.8580, both CLEAN. Universal L2 framing.	**CONFIRMED**
C040	Large-N CV L2 (CO labeling, $N = 500$/class, 5-fold): CV = 0.7629 ± 0.0120. All folds CLEAN.	**CONFIRMED**
C042	L1 INVERTED_U across training stages: BASE = 0.740, SFT = 0.803, Reasoning = 0.725	SUPPORTED
C043	L2 MONOTONE_RISE across training stages: gap BASE = 0.064, SFT = 0.096, Reasoning = 0.148	SUPPORTED
C045	Qwen3-1.7B (non-R1 lineage): commit_pct = 99.8%. Teacher-independent. Regularity 2 holds in 2 lineages.	SUPPORTED

Table 23: Experiment registry summary.

ID	Summary	Status
EXP_T1A_LARGE_N_V2	Large-N L1: Qwen = 0.7312 CI=[0.63,0.83] $N = 197$; Llama = 0.7464 CI=[0.65,0.83] $N = 200$. Both CLEAN.	COMPLETE
EXP_FALSE_CERTAINTY_V2	L2 Fisher vs. entropy, entropy-matched CC/CW (Qwen). Fisher = 0.854, gap = +0.240.	COMPLETE
EXP_FALSE_CERTAINTY_LLAMA	L2 Llama replication: Fisher = 0.818, gap = 0.365, BO_Transfer = 0.768.	COMPLETE
EXP_J_PERTURBATION_V1	Perturbation battery Qwen: ICC = 0.913, between/within = 10.5:1. C025 initial.	COMPLETE
EXP_J_PERTURBATION_V2	Perturbation battery Llama: ICC = 0.9334, between/within = 14.0:1. C025 **CONFIRMED**.	COMPLETE
EXP_L2_LARGE_N_V1	L2 cross-validated: CV = 0.7629±0.0120, $N = 500$/class, 5-fold. C040 **CONFIRMED**.	COMPLETE
EXP_OOD_GENERALIZATION_V2	OOD: HotpotQA = 84.5% efficiency; MMLU ≈ random. C041 format-scope.	COMPLETE
EXP_L_STAGE_SWEEP_V2	Stage sweep matched N: L1 INVERTED-U (C042), L2 MONOTONE-RISE (C043).	COMPLETE
EXP_CO_GEMMA_MISTRAL_V1	CO labeling on T2_L2 archs: Gemma = 0.8368, Mistral = 0.8580. C036 **CONFIRMED**.	COMPLETE
EXP_PHI_BILATERAL_V1	Phi-3.5-Mini: L1 = 0.8456 (highest), L2 = 0.7120, gap = 0.1216. C044 SUPPORTED.	COMPLETE
EXP_TEACHER_INDEPENDENCE_V1	Qwen3-1.7B: commit% = 99.8% (commits within 1–2 tokens). Regularity 2 teacher-independent. C045.	COMPLETE
EXP_MATH_AUROC_V2	MATH N=100/class: AUROC = 0.8558 CI=[0.729,0.957], L25=L26. C039 corrected.	COMPLETE
EXP_REASONING_GEOMETRY_V1	DeepSeek-R1-Qwen commit% = 75.8% (clean shuffled; EXP-B).	COMPLETE
EXP_REASONING_GEOMETRY_LLAMA	DeepSeek-R1-Llama commit% = 82.9% (clean shuffled; EXP-C).	COMPLETE
EXP_G_REASONING_ENTROPY	Post-</think> BURST: peak = 0.6932, traj = 0.8424 (GSM8K).	COMPLETE
EXP_I_EARLY_EXIT_V2	Early exit causal ($N = 500$): Δf1 = +0.012, 90% savings, $t = 6.0$, $p < 0.0001$. C028 **CONFIRMED**.	COMPLETE
EXP_H_CC_CW_PATCHING	Detection axis dissociable from answer-selection: max ΔF1 = +0.0004 under centroid patching. C024.	COMPLETE
EXP_F_COMMIT_QUALITY	Commit-point CC/CW: AUROC = 0.650, BLIND. C023 EXPLORATORY.	COMPLETE
EXP_SCALE_EXTENSION_V1	7B-Instruct: Fisher = 0.8402, Entropy = 0.9645 (+0.11). C029.	COMPLETE
EXP_K_PYTHIA_LARGE_N_V4	Pythia: C026 CTX_DEP = 0 on base LMs.	STOPPED
EXP_L_STAGE_SWEEP_V1	Qwen base bilateral oracle applicable. C030 EXPLORATORY.	COMPLETE